Classification of EEG Signals Using a Common Spatial Pattern Based Motor-Imagery for a Lower-limb Rehabilitation Exoskeleton

Chih-Jer Lin
Graduate Institute of Automation Technology
National Taipei University of Technology
Taipei, Taiwan
cjlin@mail.ntut.edu.tw

Chia-Hui Lin
Graduate Institute of Automation Technology
National Taipei University of Technology
Taipei, Taiwan

Abstract*—This article presents a lower-limb exoskeleton rehabilitation integrated with an electroencephalogram (EEG) based brain-computer interface (BCI). The BCI is used to record signals collected using the EEG interface during resting, a motor imagery (MI) and a motor execution (ME) tasks. The MI data is generated when a subject imagines the movement of a limb. Therefore, the signals were characterized by Common Spatial Pattern (CSP), Power Spectrum Density (PSD) and Discrete Wavelet Transform (DWT) in the specific Mu band of Auto Regressive model (AR). In the case of the lower-limb representation, there is a problem of reliably distinguishing leg movement intentions. The study shows how the combined use of multi-model signals can improve the accuracy and reliability of the human-machine interface. The signals induced by CSP, PSD, and DWT+AR are used for the lower-limb exoskeleton control commands to drive the movement in real time. With using the support vector machine (SVM), the signals are tested the classification precision. The classification of the control system permits one to drive the lower limb rehabilitation exoskeleton effectively.

Keywords: EEG, BCI, MI, CSP, DWT, PSD, lower limb rehabilitation.

I. INTRODUCTION

The Brain Computer Interface (BCI) has become the communication system between the users and the rehabilitation devices [1-12]. The BCI controller acts as a bridge between the user and the rehabilitation devices. The data that BCI collects impacts how we collectively navigate a new digital age likely. However, the BCIs have been designed to use EEG signals in a wide variety of ways for control, especially for severely disabled subjects who suffer from motion disorder such as amyotrophic lateral sclerosis (ALS) and spinal cord injury (SCI) [2,4,6]. The motor imagery (MI) based BCIs, where users imagine movements occurring in their limbs to control the system, have been subject to extensive researches [3-7]. The BCI system can not only help the patient control objects, but also increase the neural connections between the brain and limbs as a way to replace or restore impaired communication or motor function.

The BCI is used to create a muscle-free channel to provide communication. It is used to detect whether the patient has an intention to move or not; then, it converts control commands immediately to the output devices. The system architecture displays the purpose of the controller, when the brain signals are detected from the human scalp. The MI refers to the use of the brain to imagine physical movements without actual physical behavior and the controller completes the subsequent actual movement without external stimulation [1-12]. When the person imagines the movement state, the brain waves can present the corresponding brain waveform. In the past research, many studies will be presented through the visual interface to reduce the difficulty of motor imagination. Bringing open-source principles of adaptability and transparency to BCI tools can help democratize the technology.

The lower limb exoskeletons are powerful instruments for the clinical rehabilitation of the patients with impaired lower-limb function. However, controlling the exoskeletons requires a physical action typically. It is quite different from how normal motor actions are initiated and a more intuitive way of interacting with exoskeletons is to use endogenous brain signals. This can be implemented using BCI based on EEG signals generated independently from external stimulation; it allows the movement to be fully controlled by a subject. A typical example using BCI is a system based on sensorimotor rhythms such as MI [7-9]. The BCI-mediated lower limb movement can accelerate in rehabilitation of patients by promoting neural detection, which can be considered a promising technique to accelerate rehabilitation.

Although some studies have demonstrated the efficiency of the BCI-assisted therapy of lower-limb [6-8], there is few research on patients with lower limb impairments using the EEG for classification. The foot MI-based BCIs create difficulties in the real-time implementation of lower-limb exoskeleton control. The lower-limb BCIs for exoskeleton operation is considered difficult, because the EEG signal is usually impaired by intensive body movements and tonic muscle activity. In this paper, we propose a novel human machine interface with the EEG; the CSP, PSD, and DWT+AR are used to identify the lower-limb exoskeleton control commands. The experiment, which led to development of the technique, was conducted in eight healthy subjects. The results show that the CSP modalities can improve the reliability of movement prediction by the positive detection rate of EEG-based classifications.

II. SYSTEM SETUP

The lower-limb exoskeleton system based on the BCI-based controller consists of the following parts: (1) an EEG signal recording module, (2) EEG signal processing and classifier module, (3) the control system of the lower-limb exoskeleton, and (4) a lower-limb exoskeleton rehabilitation system. The signal recording module recorded the EEG signals during the MI by subjects. Then, the EEG data were sent to the processing and classification module, in which the raw data were preprocessed by the feature extraction. To recognize the motion intention, the preprocessed data are analyzed by the classifier. After the motion intention is predicted, then the corresponding commands are sent to the control system for the execution. The block diagram of the EEG-based lower-limb exoskeleton is given in Fig. 1.

IEEE EUROCON 2023 - 20th International Conference on Smart Technologies | 978-1-6654-6397-3/23/$31.00 ©2023 IEEE | DOI: 10.1109/EUROCON56442.2023.10198960

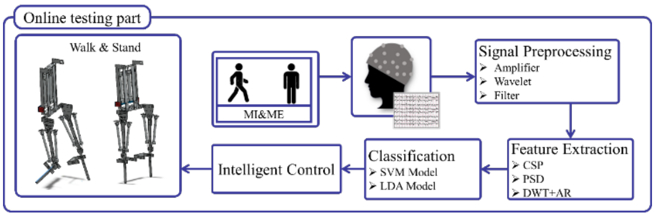


Fig. 1. EEG-based lower-limb exoskeleton

A. EEG signal acquisting module

The EEG signals were recorded using an 8-channel electroencephalography amplifier for the brainwave signal acquisition (Hippo Screen Neurotech Corp, "EAmp-0001", Taiwan). The brainwave signals were acquired using non-invasive electrodes and the original brainwave signals were amplified and converted into digital signals by transmission to the computer. Eight electrodes were used to record EEG (FP_1 , FP_2, C_3, C_z, C_4, CP_3, CP_z, CP_4) arranged according to the international 10-20 scheme (Fig. 2) [2,5-12]. The reference electrode was placed on the earlobe. The signal sampling rate was 500 Hz. Reference under the electrodes did not exceed 10 KΩ. The brain wave signals were collected from the subjects during the resting and motor imagery tasks. In the motor imagery procedure, participants were asked to follow visual commands to increase imagery in each experiment for the rest and walking tasks.

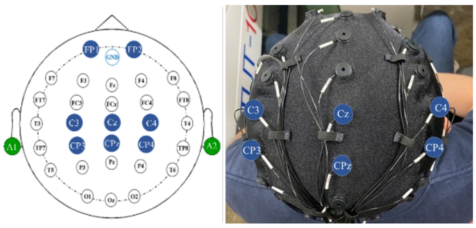


Fig. 2. Eight electrodes for the experiments

B. Lower-limb exoskeleton rehabilitation

The lower-limb exoskeleton rehabilitation system (LLERS) was designed by the National Taipei University of Technology to help disables with rehabilitation training and walking assistance. The user stands on a treadmill, which is actuated by a motor (Smart motor, SM34165DT, made by Moog Animatic). The smart motor system is integrated with a motor, an encoder, an amplifier and a controller, which is communicated with the PC via RS-232/RS-485 interface. The exoskeleton is composed of the mechanical body consists of eight pneumatic artificial muscles (PAM) to actuate. As shown in Fig. 3, the mechanism design of the PAM-actuated exoskeleton (PAMAE) has eight degrees of freedom for gait training tasks. The PAMAE's knee and hip joints are both attached to pulleys, which are actuated by the PAMs. Figure 3(a) shows that a similar design is applied to the hip joint. Because the loading of the hip is larger than that of the knee, the hip mechanism is actuated by a larger PAMs (MAS-40-300N-AA-MC-O-ER-BG). On the knee joint, the pulley is connected to a steel wire whose two ends are pulled by PAMs (MAS-20-300N-AA-MC-O-ER-BG). The potentiometers with high resolution are used to measure the joint angular positions of the hip and knee of the PAMAE; then, the positioning signals are fed back to the embedded control. To provide a walking rehabilitation task, the proposed PAMAE system is integrated with a treadmill system for gait motion training as shown in Fig. 3(b). The whole system is a highly programmable integrated system and the control unit commands the LLERS to perform the required movements and the exoskeleton can perform movements such as various types of walking. Motion initiation can be performed by control signals arising from the EEG signal classifier. For safety reasons, the control system can always be intercepted if the cases of incorrect classification appears.

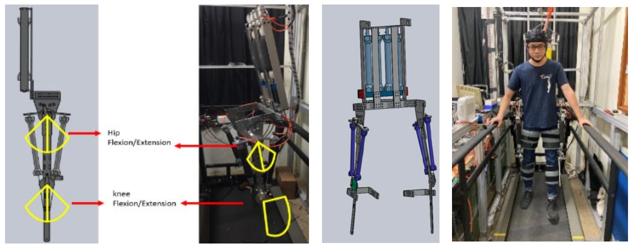


Fig. 3.(a)DOFs of the PAMAE; (b) Laboratory setup.

III. METHOD

A. Data Preprocessing

The raw EEG data were filtered by the IIR bandpass filter with frequency ranges from 2 to 40 Hz. Each session has 5 trails, the experimental paradigm is shown in Fig. 4. A trail consists of three parts: 5s for rest, 2s for the arrow cue and preparation, 5s for the motor imagery [2], [6]-[7], [9].

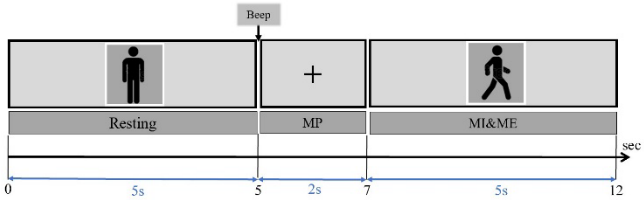


Fig. 4. MI Interface Tasks

B. Signal Processing

In this study, we used a brainwave acquisition device to collect 8 electrode EEG signals from 8 subjects (23 years ± 2 years old), the window size of the EEG signal data was set to a sampling frequency of 500 Hz, and the window offset was set to a buffer window of 300 milliseconds to capture the data. The EEG data was sent to MATLAB (2021b, MathWorks) in an array size of 150 x 8 for each transmission. The original data were processed by applying 3rd order infinite impulse response (IIR) [5] band pass(2-40Hz) and band stop(50-60Hz) filters. The signal was removed from the 60 Hz lines noise.

C. Feature Extraction

In terms of feature extraction, the best feature is used to determine the accuracy of the system with reference and the proposed method refers to specific frequency bands including as follows.

- Common Spatial Pattern (CSP)

The common spatial pattern (CSP) [1],[3]-[4],[7]-[8] filter is sourced from a preprocessing technique (filter), which is used to separate a multivariate signal into subcomponents that have maximum differences in variance. The difference allows the simple signal classification. Generally, the filter can be described as a spatial coefficient matrix W as shown in Eq. (1):

$$s = W^T E \qquad (1)$$

,where E is the multichannel EEG signals of $M \times N$ dimension, s represents signal decomposition components (matrix of $M \times N$ dimension), and W is a decomposition matrix of $M \times M$ dimension. The decomposition matrix W is calculated according to the following. An intraclass average covariance matrix for the EEG signals from classes 1 and 2 is calculated as follows.

$$C_i = \frac{1}{N_i}\sum_{l=1}^{N_i} C_l \, , i = 1,2 \tag{2}$$

, where i denotes class 1 or 2, N_i is the number of EEG signal from class i , and C_l is the covariance matrix for signal E. The average covariance matrix for the entire data is as follows.

$$C_M = C_1 + C_2 \tag{3}$$

The eigenvalues λ and the eigenvectors U of C_M can be obtained as follows.

$$C_M U = U\,\lambda \tag{4}$$

The whitening transformation matrix is P as follows.

$$P = \lambda^{-\frac{1}{2}} U^T \tag{5}$$

The matrix C_1 is transformed into

$$PC_1P^T = K \tag{6}$$

Diagonalization of matrix K gives

$$K = U_K^T \lambda_K U_K \tag{7}$$

, where λ_K is the diagonal matrix of the eigenvalues and U_K is eigenvectors. Then, the decomposition matrix W must satisfy

$$\mathrm{W}^T \mathrm{C}_1 W = \lambda_1,\ \mathrm{W}^T \mathrm{C}_2 W = \lambda_2,\ \ \lambda_1 + \lambda_2 = I \tag{8}$$

, where λ_1 and λ_2 are the diagonal matrixes of the eigenvalues. The resulting decomposition matrix W is determined as follows.

$$W = P^T U_K \tag{9}$$

The maximum dispersion values for the signal $s = W^T E$ will be observed in the first k channels.

- Discrete Wavelet Transform (DWT)

DWT [3][10] is the use of the original signal through the high-pass and low-pass filters. It is divided into an approximate signal and a detailed signal and the signal of a specific Mu band (8-12Hz) is extracted, as shown in Eq.(10)-(11).

$$\varphi_{j,k}[t] = 2^{\frac{j}{2}} \sum_k d_{j,k} \varphi[2^j t - k] \tag{10}$$

$$\psi_{j,k}[t] = 2^{\frac{j}{2}} \sum_k c_{j,k} \psi[2^j t - k] \tag{11}$$

- Power Spectrum Density (PSD)

PSD [3]-[4],[13] is the power spectrum density of the original band, which is multiplied by an appropriate coefficient to obtain the power carried per unit frequency wave. The power spectrum density of a signal exists only when the signal is in a steady process. If the signal is not in a steady state, then the autocorrelation function generates a function of two variables and the power spectrum density does not exist in such a state, as shown in Eq. (12):

$$f_\xi(t) = E\big(f(t)\big) = \lim_{T\to\infty} \frac{E[|F_T(\omega)|^2]}{T} (rad^2/Hz) \tag{12}$$

- Auto Regressive Model (AR)

AR [3][10] is a model that linearly combines the random variables of a prior time to describe the random variables of a later time, and is essentially a linear forecast, as shown in Eq. (13).

$$X_t = c + \sum_{i-1}^{p} \varphi_i X_{t-i} + \varepsilon_t \tag{13}$$

, where c is a constant term, p denotes the order of the AR model, and ε_t is the random error value with variance σ^2, mean equal to 0, and standard deviation equal to σ ; ε_t is assumed to be constant for any t. The EEG signals are collected with the brainwave signal acquisition (EAmp-0001) and then the signals are amplified and filtered by the infinite impulse response (IIR) filter. Two methods are used to capture the characteristic patterns and then the SVM is used to identify the intuition using the training data.

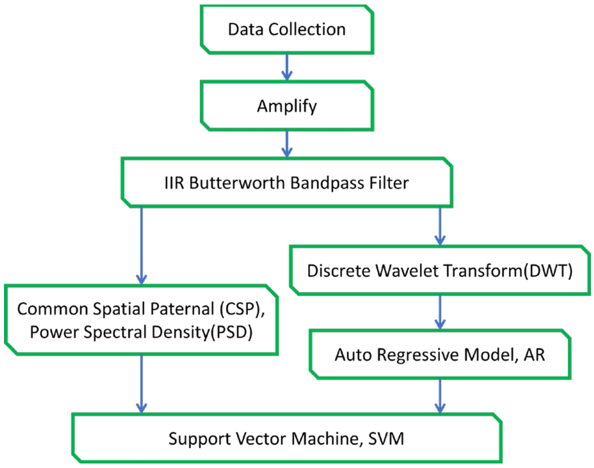


Fig. 5. EEG signals captured for classification by CSP, PSD, DWT.

D. Classification

After the EEG signals were collected and filtered by the IIR filter, the support vector machine (SVM) was used to classify the signals and the controller was used to perform the rehabiltiation task as shown in Fig. 6.

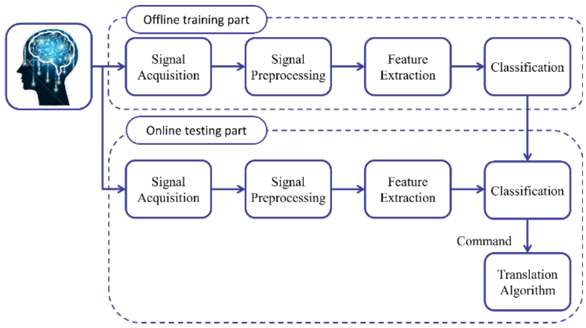


Fig. 6. Block diagram for EEG signals used to preform rehabilitation.

The support vector machine (SVM) classifier is used to maximize the distance between the separating hyper plane and the nearest training points via the termed support vectors [3]-[4],[10]. The separating hyper plane in the 2D feature space is given by the Eq (14):

$$y = \omega^T x + b \tag{14}$$

, where ω, $x \in R^2$ and the scalar $b \in R^1$; ω is the hyperplane vector, x is the support vectors and b is called a bias. The hyperplane (also called the decision border) divides the feature space into two classes. In the SVM, the distances between the hyperplanes and the nearest examples are called margins. To maximize the distance between two hyperplanes, the $\|\vec{w}\|$ should be minimized. Consider the problem of binary classification and training data are given as follows.

$$(\mathbf{x}1, y1), (\mathbf{x}2, y2), \ldots, (\mathbf{x}l, yl),\ \mathbf{x} \in R^2,\ y \in \{+1,-1\} \tag{15}$$

, where a label with the value of +1 denotes that the vector is classified to class +1 and a label of -1 denotes that the vector is part of class -1. By using given training examples, during the learning stage, the SVM finds parameters $\omega = [\omega 1, \omega 2, \ldots, \omega n]^T$ and b of a discriminant or decision function $d(x, \omega, b)$ given as

$$d(x, \omega, b) = \omega^T x + b \qquad (16)$$

In cases of linearly separable data, the boundary is also a (separating) hyperplane but of a lower order than $d(x, \omega, b)$. The decision (separation) boundary is an intersection of a decision function $d(x, \omega, b)$ and a space of input features. It is given by $d(x, \omega, b) = 0$. Therefore, the EEG signals are classified into two type using the SVM.

IV. Case Study

In the offline training mode, the brain waves are extracted from the original data and the features are put into the SVM model. Then, the EEG data are separated into two cases by the classifier. The data consists of eight bipolar recordings (FP_1, FP_2, C_3, C_z, C_4, CP_3, CP_z, CP_4) corresponding to the image (shown in Fig. 2). During the acquisition process, subjects under study were told to imagine left leg or right leg movement for five seconds after the cue was initialized, roughly two seconds after the arrow cue appears. Imagination of left or right leg's movement depended on the movie shown in the screen. Fig. 7 shows an example that three scheme paradigm appears and the EEG signals are captured. In this case study, there are eight subjects participated in the training sessions and each session contained three runs with trails and two MI classes. A total 40 trials per session were acquired, leading to 40 repetitions of left MI class and 40 repetitions of right MI class out of the two sessions. These data set are all used in this case study.

Fig. 7. Three scheme paradigms for capturing the EEG signals.

A. Data Processing workflow

The main purpose of the data processing was to compare the different extraction techniques under the same conditions. The overall methodology followed for the techniques was shown in Fig. 6. In the online test mode, brain waves are captured from the original data, the features are calculated and then classified by the SVM classifier and the output features are converted into control commands corresponding to the categories. The identified commands are transmitted by TCP/IP to the LabVIEW CompactRIO controller to produce the corresponding commands to the lower limb rehabilitation exoskeleton system.

B. Data Structure to Verification

In the implementation, there are eight subjects to perform the experiments and their EEG signals are captured and the features are classified by the CSP, DSP and the DWT+AR methods. To classify the EEG data via the SVM, the classification tests are divided into three types, which are the internal training data, half of internal training data, and all external test.

- All Internal test

The training data is used to the classifier for training. Then, the same training data is used as test data to the classifier for testing.

- Half of Internal

The training data is used to the classifier for training. Then, it take half of the training data randomly as test data. The half of the test data is chosen randomly as test data to the classifier for testing.

- All External

The training data is taken to the classifier for training. Then, the other test data is used to the classifier for testing.

To compare from the difference performances of the CSP, PSD and DWT+AR. The experimental data performed by the eight subjects are used to compare their differences of the three methods. Tables 1-3 shows the offline training mode of data structure for all the subjects' data using CSP, PSD and DWT+AR methods for the individual data. From the results in Tables 1-3, it shows that the classification rate of the CSP is the best of the three methods. The results show that the CSP has the best classification rate for all internal test; the best one is almost 90%. In the PSD case study, the best one of all external tests is almost 74%. However, the classification rate of PSD is almost 66.9% for all internal test; the classification rate of DWT+AR is almost 61.3% for all internal test. In this case study, the CSP shows its better performance of classification than the PSD and DWT+AR.

TABLE I. CSP Feature Classification Validation Chart

	All Internal Test	Half of Internal Test	All External Test
S1	82.7%	76.7%	68.7%
S2	84.0%	74.0%	67.3%
S3	90.7%	84.7%	74.0%
S4	83.3%	82.7%	73.3%
S5	80.7%	72.0%	64.0%
S6	83.3%	72.7%	68.0%
S7	80.7%	76.0%	66.7%
S8	79.3%	76.0%	71.3%

TABLE II. PSD Feature Classification Validation Chart

	All Internal Test	Half of Internal Test	All External Test
S1	65.3%	60.0%	48.7%
S2	69.3%	57.3%	50.0%
S3	62.7%	57.3%	50.0%
S4	74.7%	68.7%	57.3%
S5	63.3%	54.0%	50.0%
S6	65.3%	58.7%	50.0%
S7	68.7%	66.6%	50.0%
S8	66.0%	53.3%	50.0%

TABLE III. DWT+AR Feature Classification Validation Chart

	All Internal Test	Half of Internal Test	All External Test
S1	58.0%	56.7%	47.3%
S2	60.7%	60.0%	52.0%
S3	62.0%	57.3%	59.3%
S4	66.0%	64.0%	60.7%
S5	61.3%	54.7%	52.7%
S6	64.0%	60.7%	55.3%
S7	58.0%	57.3%	54.0%
S8	60.7%	57.3%	55.3%

To check the classification features of the CSP, Fig. 8 shows that the subjects S1 to S8 have concentrated feature distributions in Fig. 8(a)-(h). From the experimental results, the subject S3 has the best situation to produce the classification rate in Fig. 8(c) as the results in Table 1. To check the classification results of the PSD, Fig. 9 shows some experimental results of the subjects S2, S3, S4, and S6. From Fig. 9 (b) and (d), the classification results of subjects S3 and S6 have diffusion feature distribution, so the accuracy of the classifier is reduced if multiple data are put into the classifier. From Fig. 10, it shows the classification results of S1, S3, S4 and S7 for the DWT+AR. The result of S4 shows the best classification rate of the DWT+AR tests. Form the PSD and DWT+AR characteristics, we can see that the band distribution of the classification is different and it leads to a significant decrease in the accuracy of the whole classifier. Therefore, if the data of all subjects are mixed, the classifier prediction will be confused and there is no accurate database for testing. Therefore, each subject needs to be trained with customized characteristics.

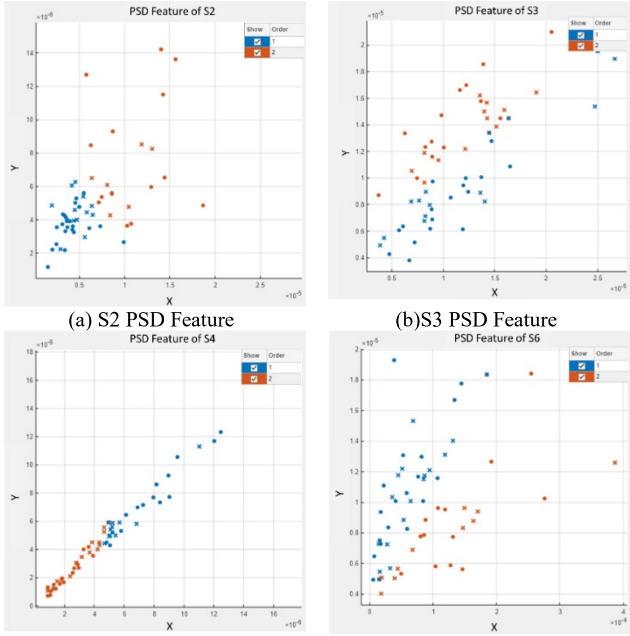
(a) S1 CSP Feature (b)S2 CSP Feature

(c)S3 CSP Feature (d) S4 CSP Feature

(e)S5 CSP Feature (f) S6 CSP Feature

(g)S7 CSP Feature (h)S8 CSP Feature

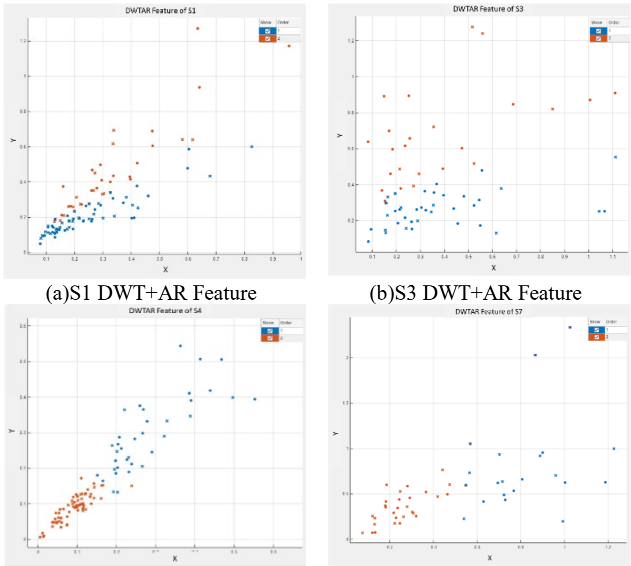
Fig. 8 Distribution of the CSP features



(a) S2 PSD Feature (b)S3 PSD Feature

(c)S4 PSD Feature (d) S6 PSD Feature

Fig. 9 Distribution of the PSD features



(a)S1 DWT+AR Feature (b)S3 DWT+AR Feature

(c) S4 DWT+AR Feature (d)S7 DWT+AR Feature

Fig. 10 Distribution of the DWT+AR features

When the data in the off-line training mode are compared, it is found that the accuracy rate of fully internal data is higher than that of the semi-internal and external data structures. Based on the online test, it was found that the CSP has the best classifier rate than the other two methods. Therefore, the CSP can be used for the rehabilitation machine to perform the motion execution. However, the PSD and DWT+AR methods could not be used for the online prediction test due to the low accuracy. The subjects could not make obvious changes to the feature when the tasks of rest state and motion imagination state appear. If the mean value of the EEG signals is not significantly changed within the limited time, the feature cannot be recognized with using the motor imagination. Therefore, the feature cannot be identified to drive the lower limb rehabilitation machine in this experiment.

C. Verification Method

The above classification features are programmed in the classifier for detecting and classifying the task patterns. To evaluate the accuracy of the system, the feature database is partitioned into a training database and a test database using the k-Fold validation technique. The training data is used to construct the classifier, while the test data is used for evaluation. In the k-fold cross-validation, the training set is partitioned into k subsamples. A single subsample is kept as data for validating the model and the other k-1 samples are used for training [1],[7],[10-15]. The cross-validation is repeated k times for each subsample and the results are averaged over k times or combined in other way to obtain a single estimate. In experiments with motor execution, the classification results for the tests using the CSP, PSD, and DWT+AR methods are shown in Tables 1-3. From the results, the CSP method has the best classification accuracy for three cases. Then, the LLERS is integrated with the EEG system to be implemented in clinical rehabilitation protocols for robotic exoskeleton as Fig. 11. Figure 12 shows an example for the implementation of the whole system.

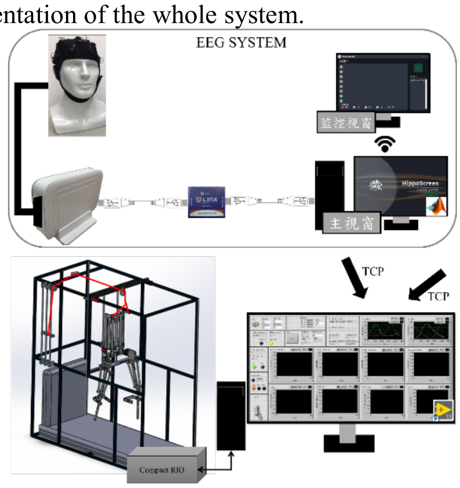


Fig. 11. LLERS integrated with the EEG system.

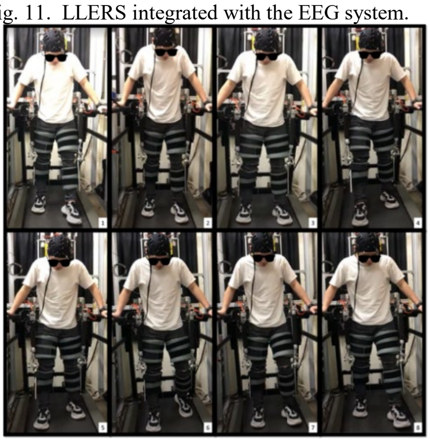


Fig. 12. Example for the implementation for the proposed system.

V. Conclusion

In this study, the EEG signals were acquired from eight subjects (23 ± 2 years old) for motor imagery tasks at the rest and walling. With using experimental data from all subjects trained in all internal data, the experimental data from individual subjects tested with the classification of the CSP, PSD and DWT+AR. In the case studies of these three methods, the CSP has the best classifier rate than the other two methods. The results show that the best classification rate of the CSP is almost 90%. However, the best classification rate of PSD is almost 66.9% for all internal test; the best one of DWT+AR is almost 61.3% for all internal test. Furthermore, the EEG signals can be further implemented in clinical rehabilitation of robotic exoskeletons, neuromuscular disorder, and state of rehabilitation therapy.

Acknowledgment

This research was funded by National Science Council of the Republic of China, grant number Contract No. MOST 108-2221-E-027-112-MY3. The APC was funded by Contract No. MOST 111-2221-E-027-146-MY3.

108-2221-E-027-112-MY3.References

[1] E. S. Kang, B. Kim, H. Suk, "An empirical suggestion for collaborative learning in motor imagery-based BCIs," *2016 4th International Winter Conference on Brain-Computer Interface (BCI)*, 1-3, 2016, doi: 10.1109/IWW-BCI.2016.7457450.

[2] G. Yu, J. Wang, W. Chen and J. Zhang, "EEG-based brain-controlled lower extremity exoskeleton rehabilitation robot," *2017 IEEE nternational Conference on Cybernetics and Intelligent Systems (CIS) and IEEE Conference on Robotics, Automation and Mechatronics (RAM)*, 763-767, 2017, doi: 10.1109/ICCIS.2017.8274875.

[3] M. Tariq, P.M. Trivailo, M. Simic, "EEG-Based BCI Control Schemes for Lower-Limb Assistive-Robots. Front Hum Neurosci. 312, 2018, doi: 10.3389/fnhum.2018.00312.

[4] N. Padfield, J. Zabalza, H. Zhao, V. Masero, J. Ren, "EEG-Based Brain-Computer Interfaces Using Motor-Imagery: Techniques and Challenges," *Sensors,* 19(6):1423, 2019, doi:10.3390/s19061423.

[5] J. Choi, H. Kim, "Real-time Decoding of EEG Gait Intention for Controlling a Lower-limb Exoskeleton System," *2019 7th International Winter Conference on Brain-Computer Interface (BCI)*, 1-3, 2019, doi: 10.1109/IWW-BCI.2019.8737311.

[6] G. Roy, D. Nirola, S. Bhaumik, "An Approach towards Development of Brain Controlled Lower Limb Exoskeleton for Mobility Regeneration," *2019 IEEE Region 10 Symposium (TENSYMP)*, 385-390, 2019, doi: 10.1109/TENSYMP46218.2019.8971173.

[7] S.Y. Gordleeva, S.A. Lobov, N.A. Grigorev, A.O. Savosenkov, M.O. Shamshin, M.V. Lukoyanov, M.A. Khoruzhko, V.B. Kazantev, "Real-Time EEG–EMG Human–Machine Interface-Based Control System for a Lower-Limb Exoskeleton," *IEEE Access*, 8, 84070-84081, 2020, doi: 10.1109/ACCESS.2020.2991812.

[8] J. Choi, K. -T. Kim, J. Lee, S. J. Lee, H. Kim, "Robust Semi-synchronous BCI Controller for Brain-Actuated Exoskeleton System," *2020 8th International Winter Conference on Brain-Computer Interface (BCI)*, 1-3, 2020, doi: 10.1109/BCI48061.2020.9061658.

[9] N.A. Grigorev , A.O. Savosenkov, M.V. Lukoyanov, A. Udoratina, N. N. Shusharina, A.Ya. Kaplan, A.E. Hramov , V.B. Kazantsev, S. Gordleev, "A BCI-Based Vibrotactile Neurofeedback Training Improves Motor Cortical Excitability During Motor Imagery," IEEE Transactions on Neural Systems and Rehabilitation Engineering, vol. 29, pp. 1583-1592, 2021, doi: 10.1109/TNSRE.2021.3102304.

[10] Y. Zhang, B. Liu, X. Ji, "Classification of EEG Signals Based on Autoregressive Model and Wavelet Packet Decomposition," *Neural Process Lett* 45, 2017, 365–378.

[11] D. Liu, W.Chen, Z. Pei, J. Wang, "A brain-controlled lower-limb exoskeleton for human gait training," Rev Sci Instrum. 2017 Oct;88(10):104302. doi: 10.1063/1.5006461. PMID: 29092520.

[12] M.S. Al-Quraishi, I. Elamvazuthi, S.A. Daud, S. Parasuraman, A. Borboni "EEG-Based Control for Upper and Lower Limb Exoskeletons and Prostheses: A Systematic Review," *Sensors*, 18(10), 3342, 2018, doi: 10.3390/s18103342.

[13] P. Li, X. Wang, F. Li, R. Zhang, T. Ma, Y. Peng, X. Lei, Y. Tian, D. Guo, T. Liu, D. Yao, P. Xu, "Autoregressive model in the Lp norm space for EEG analysis," Journal of Neuroscience Methods,Vol 240, 170-178, 2015, ISSN 0165-0270.

[14] Y. Zhang, S. Zhang, X. Ji, "EEG-based classification of emotions using empirical mode decomposition and autoregressive model," *Multimed Tools Appl* 77, 26697–26710, 2018.

[15] Y. Guo, R. Gravina, X. Gu, G. Fortino, G.Z. Yang, "Emg-based abnormal gait detection and recognition," 2020 IEEE International Conference on Human-Machine Systems (ICHMS)